Locality from Hidden Zeros: Cyclic 1-zeros uniquely determine ϕ^3 tree amplitudes

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Abstract

We prove that the cyclic 1-zero conditions of Rodina uniquely determine the ordered ϕ^3 tree amplitude $\mathcal{A}_n^{\text{tree}}$ within the full space of rational functions with simple poles in planar Mandelstam variables—a space that includes all possible non-local (crossing-chord) contributions. At $n = 5$ and $n = 6$ the proof is carried out by explicit symbolic computation: every non-local term is eliminated by the 1-zeros, and the tree amplitude emerges as the unique survivor. At $n = 7$ the result is confirmed numerically. In all cases, *locality* (poles only in planar channels) is a *derived consequence* of the 1-zero conditions, not an assumption. This is in contrast to Rodina's original proof, which works within the space of Feynman-diagram sums and therefore assumes locality from the outset.

Contents

1 Introduction **1**
1.1 Main results . . . 1

2 Preliminaries **2**
2.1 Planar Mandelstam variables . . . 2
2.2 The 1-zero loci . . . 2
2.3 The ansatz space . . . 2
2.4 The tree amplitude . . . 2

3 Part I: The $n = 5$ base case **2**
3.1 Setup . . . 2
3.2 Leg-by-leg analysis . . . 3
3.3 The theorem at $n = 5$ . . . 4

4 Part II: The $n = 6$ case **4**
4.1 Setup . . . 4
4.2 Leg-by-leg rank analysis . . . 4
4.3 Structural features . . . 5

5 The $n = 7$ case (numerical) **5**

6 Discussion **5**
6.1 Summary of results . . . 5
6.2 Locality is derived . . . 5
6.3 The conjecture . . . 6
6.4 Relation to Rodina's proof . . . 6
6.5 Toward a general proof . . . 6

1 Introduction

Rodina [1] proved that ordered tree-level amplitudes in $\mathrm{Tr}(\phi^3)$ theory are uniquely characterised by their *hidden zeros*: the amplitude $\mathcal{A}_n^{\text{tree}}$ vanishes on each of the n cyclic 1-zero loci $\mathcal{Z}_1, \ldots, \mathcal{Z}_n$ in the space of planar Mandelstam variables. Rodina's proof works within the space of local (Feynman-diagram) ansatze and establishes that the 1-zeros fix all relative coefficients—i.e., they prove *unitarity* given locality.

We ask the stronger question: do the 1-zeros also enforce *locality* itself? That is, if we start with the broadest possible rational ansatz—including terms with poles in non-planar (crossing-chord) channels and even higher-order poles—do the 1-zeros force all non-local terms to vanish?

We prove that the answer is yes at $n = 5$ and $n = 6$ by explicit computation, and verify it numerically at $n = 7$. The proofs are purely linear-algebraic: the 1-zero conditions impose a system of linear equations on the ansatz coefficients, and we show that the nullspace of this system is exactly 1-dimensional, spanned by the tree amplitude.

1.1 Main results

Theorem 1.1 (Informal)**.** *For $n = 5, 6, 7$, let B be a rational function of the planar Mandelstam variables, homogeneous of mass dimension $-(n-3)$, with at most simple poles. If B vanishes on each of the n cyclic 1-zero loci, then B is proportional to the ϕ^3 tree amplitude:*

$$B \;=\; c \cdot \mathcal{A}_n^{\text{tree}}.$$

In particular, every non-planar pole is eliminated by the 1-zero conditions.

The formal statements are Theorems 3.1 and 4.1 below.

2 Preliminaries

2.1 Planar Mandelstam variables

Let $p_1, \ldots, p_n$ be massless on-shell momenta satisfying $\sum p_i = 0$ and $p_i^2 = 0$. Define $X_{ij} = (p_i + p_{i+1} + \cdots + p_{j-1})^2$ for $1 \le i < j \le n$, with $X_{ij} = X_{ji}$, $X_{ii} = 0$, and $X_{i,i+1} = X_{1n} = 0$. At n points there are $d = n(n-3)/2$ independent planar variables forming the kinematic space $\mathcal{V}_n$.

2.2 The 1-zero loci

For each $k \in \{1, \ldots, n\}$, the *cyclic 1-zero at leg k* is the codimension-$(n-3)$ linear subspace $\mathcal{Z}_k \subset \mathcal{V}_n$ cut out by

$$c_{k,k+2} = c_{k,k+3} = \cdots = c_{k,k+n-2} = 0, \quad \text{where} \quad c_{ij} := X_{ij} + X_{i+1,j+1} - X_{i,j+1} - X_{i+1,j}. \tag{1}$$

2.3 The ansatz space

Definition 2.1. The *ansatz space* $\mathcal{B}_n$ is the vector space of rational functions of the $d = n(n-3)/2$ planar Mandelstam variables that are homogeneous of degree $-(n-3)$ and have at most simple poles in each variable independently. A basis for $\mathcal{B}_n$ is

$$\left\{ \frac{1}{X_{a_1} \cdots X_{a_{n-3}}} \ : \ a_1 < \cdots < a_{n-3},\ a_i \in \{X_{ij}\}_{1 \le i < j \le n} \right\},$$

so $\dim \mathcal{B}_n = \binom{d}{n-3}$.

At $n = 5$: $d = 5$, $\dim \mathcal{B}_5 = \binom{5}{2} = 10$. At $n = 6$: $d = 9$, $\dim \mathcal{B}_6 = \binom{9}{3} = 84$. At $n = 7$: $d = 14$, $\dim \mathcal{B}_7 = \binom{14}{4} = 1001$.

Remark 2.2. At $n = 5$ we also consider a broader ansatz $V_5^{(-2)}$ of dimension 15 that includes double poles $1/X_{ij}^2$. The theorem holds for both ansatze.

2.4 The tree amplitude

The ϕ^3 tree amplitude $\mathcal{A}_n^{\text{tree}} \in \mathcal{B}_n$ is the sum over all C_{n-2} (Catalan number) non-crossing triangulations of the convex n-gon:

$$\mathcal{A}_n^{\text{tree}} = \sum_{\text{triangulations}} \frac{1}{X_{a_1} \cdots X_{a_{n-3}}}.$$

Rodina's hidden-zero theorem states that $\mathcal{A}_n^{\text{tree}}$ vanishes on every $\mathcal{Z}_k$.

3 Part I: The $n = 5$ base case

3.1 Setup

At $n = 5$ the planar variables are $X_{13}, X_{14}, X_{24}, X_{25}, X_{35}$. The five 1-zero loci are listed in Table 1.

k	Substitution	Free variables
1	$X_{13} = -X_{25},\ X_{14} = X_{24} - X_{25}$	X_{24}, X_{25}, X_{35}
2	$X_{13} = X_{35} - X_{25},\ X_{24} = X_{25} - X_{35}$	X_{14}, X_{25}, X_{35}
3	$X_{13} = X_{14} + X_{35},\ X_{24} = -X_{35}$	X_{14}, X_{25}, X_{35}
4	$X_{14} = -X_{35},\ X_{24} = X_{25} - X_{35}$	X_{13}, X_{25}, X_{35}
5	$X_{13} = X_{35} - X_{25},\ X_{14} = -X_{25}$	X_{24}, X_{25}, X_{35}

Table 1: The five cyclic 1-zero loci at $n = 5$.

We work with the 15-dimensional ansatz $V_5^{(-2)}$ (including double poles), with basis $\{1/(X_{ij}X_{kl})\}_{(ij)\neq(kl)} \cup \{1/X_{ij}^2\}$ and coefficients $c_1, \ldots, c_{15}$ as in Table 2.

	term		term
c_1	$1/(X_{13}X_{14})$	c_9	$1/(X_{24}X_{35})$
c_2	$1/(X_{13}X_{24})$	c_{10}	$1/(X_{25}X_{35})$
c_3	$1/(X_{13}X_{25})$	c_{11}	$1/X_{13}^2$
c_4	$1/(X_{13}X_{35})$	c_{12}	$1/X_{14}^2$
c_5	$1/(X_{14}X_{24})$	c_{13}	$1/X_{24}^2$
c_6	$1/(X_{14}X_{25})$	c_{14}	$1/X_{25}^2$
c_7	$1/(X_{14}X_{35})$	c_{15}	$1/X_{35}^2$
c_8	$1/(X_{24}X_{25})$		

Table 2: The 15-dimensional ansatz at $n = 5$.

3.2 Leg-by-leg analysis

Leg 1. Substitute $X_{13} \to -X_{25}$, $X_{14} \to X_{24} - X_{25}$, clear denominators. Row reduction yields 9 independent equations:

$$\begin{aligned} c_1 - c_5 - c_6 &= 0, & c_4 - c_{10} &= 0, \\ c_2 + c_5 - c_8 &= 0, & c_7 &= 0, \\ c_3 - c_{11} - c_{14} &= 0, & c_9 &= 0, \\ c_{12} &= 0, & c_{13} &= 0, \\ c_{15} &= 0. \end{aligned} \tag{2}$$

After leg 1: 6-dimensional survivor space.

Leg 2. Four new independent equations:

$$c_5 - c_{10} = 0, \qquad c_6 = 0, \qquad c_8 - c_{10} - c_{11} = 0, \qquad c_{14} = 0. \tag{3}$$

After leg 2: 2-dimensional survivor space, spanned by

$$B\big|_{\text{after legs 1,2}} = c_{10}\,\mathcal{A}_5^{\text{tree}} + c_{11} \cdot \mathcal{N}, \quad \text{where} \quad \mathcal{N} = \frac{(X_{13} + X_{24})(X_{13} + X_{25})}{X_{13}^2\, X_{24}\, X_{25}}. \tag{4}$$

The non-local survivor $\mathcal{N}$ contains crossing-chord products $1/(X_{13}X_{24})$, $1/(X_{13}X_{25})$ and the squared propagator $1/X_{13}^2$.

Leg 3. Since $\mathcal{A}_5^{\text{tree}}$ vanishes on $\mathcal{Z}_3$,

$$B\big|_{\mathcal{Z}_3} = -c_{11} \cdot \frac{X_{14}(X_{14} + X_{25} + X_{35})}{X_{25}\, X_{35}\, (X_{14} + X_{35})^2} = 0 \quad \Longrightarrow \quad c_{11} = 0.$$

Legs 4, 5. Redundant: $B = c_{10}\,\mathcal{A}_5^{\text{tree}}$ already vanishes on $\mathcal{Z}_4$ and $\mathcal{Z}_5$.

3.3 The theorem at $n = 5$

Theorem 3.1. *Let $B \in V_5^{(-2)}$. If B vanishes on all five cyclic 1-zero loci, then $B = c \cdot \mathcal{A}_5^{\text{tree}}$.*

Proof. The system (2)–(3) plus $c_{11} = 0$ comprises 14 independent equations in 15 unknowns. Nullspace dimension = 1, spanned by $\mathcal{A}_5^{\text{tree}}$. □

Corollary 3.2. *Exactly 3 of the 5 cyclic 1-zeros are independent as constraints on $V_5^{(-2)}$.*

Remark 3.3 (Where non-local terms die)**.** Crossing-chord terms $c_7 = 1/(X_{14}X_{35})$ and $c_9 = 1/(X_{24}X_{35})$ die at leg 1. The crossing-chord terms c_2, c_3 and the squared propagator c_{11} survive two legs but die at leg 3. After leg 3, the surviving coefficients $c_1 = c_4 = c_5 = c_8 = c_{10}$ are exactly the 5 non-crossing triangulations of the pentagon, all with equal coefficient.

4 Part II: The $n = 6$ case

4.1 Setup

At $n = 6$ the 9 planar variables are $X_{13}, X_{14}, X_{15}, X_{24}, X_{25}, X_{26}, X_{35}, X_{36}, X_{46}$. The simple-pole ansatz $\mathcal{B}_6$ has $\binom{9}{3} = 84$ basis elements. The 6 cyclic 1-zero loci are listed in Table 3.

k	Substitution	Free variables
1	$X_{13} = -X_{26},\ X_{14} = X_{24} - X_{26},\ X_{15} = X_{25} - X_{26}$	$X_{24}, X_{25}, X_{26}, X_{35}, X_{36}, X_{46}$
2	$X_{13} = X_{36} - X_{26},\ X_{24} = X_{26} - X_{36},\ X_{25} = X_{26} + X_{35} - X_{36}$	$X_{14}, X_{15}, X_{26}, X_{35}, X_{36}, X_{46}$
3	$X_{13} = X_{14} + X_{36} - X_{46},\ X_{24} = X_{46} - X_{36},\ X_{35} = X_{36} - X_{46}$	$X_{14}, X_{15}, X_{25}, X_{26}, X_{36}, X_{46}$
4	$X_{14} = X_{15} + X_{46},\ X_{24} = X_{25} + X_{46},\ X_{35} = -X_{46}$	$X_{13}, X_{15}, X_{25}, X_{26}, X_{36}, X_{46}$
5	$X_{15} = -X_{46},\ X_{25} = X_{26} - X_{46},\ X_{35} = X_{36} - X_{46}$	$X_{13}, X_{14}, X_{24}, X_{26}, X_{36}, X_{46}$
6	$X_{13} = X_{36} - X_{26},\ X_{14} = X_{46} - X_{26},\ X_{15} = -X_{26}$	$X_{24}, X_{25}, X_{26}, X_{35}, X_{36}, X_{46}$

Table 3: The six cyclic 1-zero loci at $n = 6$.

4.2 Leg-by-leg rank analysis

Theorem 4.1. *The cumulative constraint rank evolves as follows:*

Legs imposed	*Raw eqs*	*Rank (of 84)*	*Nullspace*	*New rank*
Leg 1	63	51	33	51
Legs $1, 2$	154	76	8	25
Legs $1, 2, 3$	245	81	3	5
Legs $1, 2, 3, 4$	308	82	2	1
Legs $1, \ldots, 5$	371	83	1	1
All 6	434	83	1	0

The nullspace is 1*-dimensional, spanned by* $\mathcal{A}_6^{\text{tree}}$ *(the sum over all* $C_4 = 14$ *non-crossing triangulations of the hexagon, with equal coefficients).*

Proof. Verified by exact symbolic computation in `sympy`. At each step, the $\mathcal{Z}_k$ substitution is applied to the 84-term ansatz, denominators are cleared, and monomial coefficients in the 6 free variables are set to zero. The nullspace vector has exactly 14 nonzero entries (all equal), matching the 14 triangulations. □

Corollary 4.2. *At* $n = 6$*, exactly* 5 *of the* 6 *cyclic* 1*-zeros are independent. Leg* 6 *is redundant. The number of independent legs is* $n - 1$ *at both* $n = 5$ *and* $n = 6$*.*

4.3 Structural features

Remark 4.3 (Free-variable grading)**.** On each $\mathcal{Z}_k$, the 6 free variables split into *fully free* (3 variables not appearing in any constraint expression) and *constrained-free* (3 variables appearing in the

substitution right-hand sides). Since the fully-free variables are genuinely independent parameters, any rational function vanishing on $\mathcal{Z}_k$ can be graded by powers of these variables, and each grade must vanish independently. This is the mechanism behind leg 1's large rank contribution (51 of 84).

Remark 4.4 (Intermediate structure)**.** After 3 legs, the nullspace is 3-dimensional. Unlike at $n = 5$ (where the 2-dimensional intermediate cleanly separated into tree + non-local), the three basis vectors at $n = 6$ each *mix* tree and non-tree terms. Legs 4 and 5 contribute one new equation each, "unmixing" these three vectors to isolate the tree amplitude. This structural difference means a general-n proof cannot rely on clean tree/non-tree separation at each inductive step.

5 The $n = 7$ case (numerical)

Proposition 5.1. *At $n = 7$, the ansatz space $\mathcal{B}_7$ has $\binom{14}{4} = 1001$ basis elements. Numerical computation (SVD with multiple random seeds) gives constraint rank $= 1000$, nullspace dimension $= 1$, with the unique nullspace vector proportional to $\mathcal{A}_7^{\text{tree}}$ (the sum over all $C_5 = 42$ triangulations of the heptagon). The singular-value gap between the 1000-th and 1001-st singular values exceeds 13 orders of magnitude.*

6 Discussion

6.1 Summary of results

n	$\dim \mathcal{B}_n$	Rank	Nullspace	Method	Locality derived?
5	15	14	1	Symbolic	Yes
6	84	83	1	Symbolic	Yes
7	1001	1000	1	Numerical	Yes

6.2 Locality is derived

The central contribution of this work is showing that *locality is a consequence of the* 1*-zero conditions*, not an independent assumption. Rodina's original proof works within the space of Feynman-diagram sums (local ansatze) and proves that the 1-zeros fix all relative coefficients—establishing unitarity given locality. Our ansatz $\mathcal{B}_n$ is strictly larger: it includes every rational function with the correct mass dimension and pole structure, including crossing-chord products that cannot arise from any Feynman diagram. The 1-zeros eliminate all such non-local terms.

6.3 The conjecture

Conjecture 6.1. *For all $n \geq 5$,*

$$\dim \ker\big(\textit{all } n \textit{ cyclic 1-zeros on } \mathcal{B}_n\big) = 1,$$

with the unique generator being $\mathcal{A}_n^{\text{tree}}$.

Three consecutive base cases ($n = 5, 6, 7$) are verified. Several structural features support the conjecture:

- The free-variable grading mechanism is uniform in n: at each leg, the fully-free variables (those not entering any constraint expression) form an increasingly large independent set, and the grading kills an increasing fraction of the ansatz. At $n = 5$ one leg kills 60%; at $n = 8$ one leg kills $\sim 98\%$.
- The number of independent legs appears to be $n - 1$ at each n tested.
- The ratio (total conditions from all loci) / (needed conditions) grows superexponentially with n, so the system is massively overdetermined at large n.

A proof of Conjecture 6.1 would constitute a derivation of both locality and unitarity from the hidden-zero conditions alone.

6.4 Relation to Rodina's proof

Rodina's proof [1] uses the D-subset decomposition to organize Feynman-diagram terms by boundary-propagator structure under a BCFW shift, then proves by induction that the 1-zeros fix all coefficients within the local ansatz. The key difference with the present work:

	Rodina	This work
Ansatz space	Local (Feynman diagrams)	Full $\mathcal{B}_n$ (all poles)
Locality	Assumed	Derived
Unitarity	Derived	Derived
Method	Induction via D-subsets	Direct linear algebra
Scope	All n	$n = 5, 6, 7$ (conjectured for all n)

The two approaches are complementary: Rodina's result holds for all n but assumes locality; ours derives locality but (so far) only at specific n.

6.5 Toward a general proof

A proof of Conjecture 6.1 likely requires one of:

1. **Representation theory.** The cyclic group $\mathbb{Z}_n$ acts on both $\mathcal{B}_n$ and the constraint matrix. Decomposing by $\mathbb{Z}_n$-representations may reduce the problem.
2. **Commutative algebra.** The intersection of ideals $I(\mathcal{Z}_1) \cap \cdots \cap I(\mathcal{Z}_n)$ at the appropriate degree controls the nullspace. Hilbert function methods on arrangements of linear subspaces may apply.
3. **Combinatorics of the associahedron.** The tree amplitude is the canonical form of the kinematic associahedron. The 1-zero loci are specific faces/sections of this polytope, and the uniqueness may follow from the facial structure.

References

[1] L. Rodina, *Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in* $\mathrm{Tr}(\phi^3)$ *theory*, arXiv:2406.04234v5 (2024).